Scenario-Based Questions | Set 5

1. A language learning app wants to recommend grammar or vocabulary...

Scenario: A language learning app wants to recommend grammar or vocabulary exercises based on areas where the user struggles. How can they build this system?

Answer / Logic:

1. Track quiz scores and mistakes in grammar and vocabulary topics.

2. Identify the topics where each learner needs more practice.

3. Use content-based matching to suggest exercises for those weak topics.

4. Update the exercises as the learner improves and completes practice.

2. A smartwatch app collects data like skin temperature, heart rate,...

Scenario: A smartwatch app collects data like skin temperature, heart rate, motion, and time of day to predict stress levels. How should relevant features be selected?

Answer / Logic:

1. Clean heart rate, temperature, motion and time-of-day readings.

2. Remove missing, repeated or unreliable sensor features.

3. Use SelectKBest or Random Forest importance to find signals related to stress.

4. Check results on test data and keep the features that help prediction.

3. An online bookstore wants to recommend books similar in writing...

Scenario: An online bookstore wants to recommend books similar in writing style and genre to the ones a user previously liked. How can they achieve this?

Answer / Logic:

1. Collect book genre, descriptions and writing-style details.

2. Convert book descriptions into numerical features using TF-IDF.

3. Use cosine similarity to find books close to the ones a reader liked.

4. Recommend matching books and improve the results using reader feedback.

4. A logistics company wants to predict delivery times using features...

Scenario: A logistics company wants to predict delivery times using features like distance, traffic, weather, package weight, and route history. How can they select features?

Answer / Logic:

1. Prepare distance, traffic, weather, package weight and route history.

2. Use delivery time as the target and clean missing or unusual values.

3. Use f_regression or Random Forest importance to select useful features.

4. Compare prediction errors and keep the features that estimate delivery time well.

5. A social app wants to recommend new friends based on shared...

Scenario: A social app wants to recommend new friends based on shared interests, mutual friends, and interaction patterns. How should they implement this?

Answer / Logic:

1. Collect shared interests, mutual friends and interaction information.

2. Create user profiles or similarity scores from these details.

3. Rank possible friends using common interests and interaction patterns.

4. Remove existing friends and follow privacy settings before recommending.

6. A job portal wants to predict salary based on features like...

Scenario: A job portal wants to predict salary based on features like education, experience, industry, location, and skills. How should they reduce the feature set?

Answer / Logic:

1. Prepare education, experience, industry, location and skills as features.

2. Encode categorical details and remove duplicate or irrelevant columns.

3. Use SelectKBest with f_regression or a regression model’s feature importance.

4. Test the reduced feature set and keep the inputs that predict salary reliably.

7. A food recipe app wants to suggest dishes based on ingredients the...

Scenario: A food recipe app wants to suggest dishes based on ingredients the user has. How can they build this?

Answer / Logic:

1. Ask the user which ingredients they currently have.

2. Store ingredient lists and food tags for every recipe.

3. Use content-based matching to find recipes with the most matching ingredients.

4. Recommend suitable dishes and show any extra ingredients needed.

8. A retail chain wants to forecast product demand using past sales,...

Scenario: A retail chain wants to forecast product demand using past sales, holidays, pricing, promotion, location, and competitor activity. How can they select key features?

Answer / Logic:

1. Collect sales, holidays, prices, promotions and competitor information.

2. Create time-based features such as previous-week sales and seasonal patterns.

3. Use a forecasting model to check which features improve demand prediction.

4. Test on later dates and keep the features that give lower forecast errors.

9. A dating app wants to suggest compatible profiles based on user...

Scenario: A dating app wants to suggest compatible profiles based on user preferences, likes, and profile traits. How can they implement this?

Answer / Logic:

1. Collect profile interests, preferences and likes with user permission.

2. Build profile features and compare them using content-based matching.

3. Use past interactions to improve the ranking of suitable profiles.

4. Recommend only mutually suitable profiles while respecting privacy settings.

10. A bank wants to detect fraudulent transactions using features like...

Scenario: A bank wants to detect fraudulent transactions using features like transaction amount, time, location, device, and frequency. How should they select features?

Answer / Logic:

1. Prepare transaction amount, time, location, device and frequency data.

2. Clean missing values and create useful recent-activity features.

3. Use mutual information or tree-based importance to select fraud indicators.

4. Check precision and recall on test data before keeping the final features.